



July 9, 2016

In Insects, Viticulture

5 Comments

Japanese Beetle: A Common Pest in the Vineyard

By: Andy Muza, Penn State Extension – Erie County

Distribution

The Japanese beetle has been in the United States since 1916 when it was first discovered in New Jersey. By 1920, this pest had migrated to southeastern Pennsylvania and by 1957 this insect could be found in every county in the state. Currently, this pest can be commonly found in all states from Maine to Georgia and as far west as Illinois and Tennessee. Japanese beetle has also been found in parts of Nebraska, Kansas, Oklahoma and in Utah.

Life Cycle and Description

Japanese beetle has 1 generation/year with 4 life stages: egg, larva, pupa and adult. The beetles are almost ½" in length and ¼" wide with a metallic green body and bronze colored wing covers. An identifying characteristic is 12 patches of white hairs on the abdomen around the outside edges of the wing covers (Figure 1).



Figure 1: Adult Japanese Beetle, Photo from:
https://www.aphis.usda.gov/publications/plant_health/2015/japanese-beetle-handbook.pdf

Beetles can live for 4 – 6 weeks and they spend this time voraciously feeding and mating. Females deposit their eggs in the soil and can lay up to 60 eggs during their lifetime. Eggs hatch in 10 – 14 days and larvae (grubs) begin feeding, mainly on grass roots, near the soil surface. A fully grown larva is about 1" long with a soft, white body, 3 pairs of legs, and a light brown head capsule. Grubs are described as having a curled or C-shaped body (Figure 2). In the fall, larvae move deeper into the soil (4" – 8") to overwinter. As soil temperatures warm in the spring the mature grubs (Figure 3) return to the soil surface to feed and pupate.



Figure 2: Japanese Beetle Larva, Photo from:
https://www.aphis.usda.gov/publications/plant_health/2015/japanese-beetle-handbook.pdf

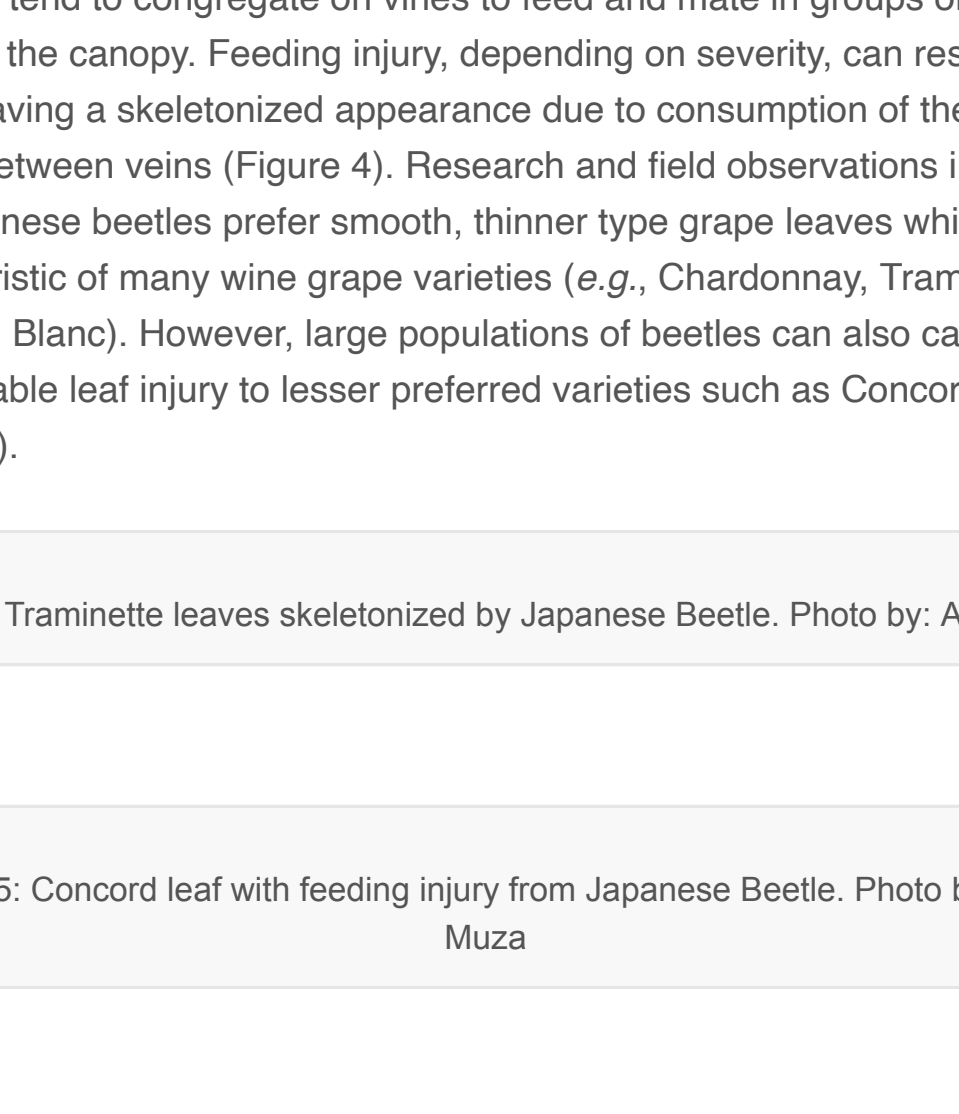


Figure 3: Mature Japanese beetle larva (grub). Photo from: S. Hesler

Emergence

On average in southern Pennsylvania, adult beetles begin emerging from the soil about the third week of June and in other areas of the state about 7-10 days later. This season, during the week of June 19 – 25, significant buildups of this beetle were reported in many orchards in southern Pennsylvania. In Erie County, Pennsylvania observed the first beetle in a vineyard on June 22 but by July 6 there was a noticeable increase in beetle numbers.

Feeding Injury

Adult Japanese beetle feed on over 300 species of plants including grape, tree fruits, small fruits, vegetables, ornamentals and various weeds. The larvae are serious pests of turfgrass. On some crops, (e.g., peaches) beetles can cause significant injury on both fruit and leaves. However, on grapes, feeding is mainly on leaf tissue. Beetles are most active on warm, sunny days and tend to congregate on vines to feed and mate in groups on the top leaves of the canopy. Feeding injury, depending on severity, can result in leaves having a skeletonized appearance due to consumption of the soft leaf tissues between veins (Figure 4). Research and field observations indicate that Japanese beetles prefer smooth, thinner type grape leaves which are characteristic of many wine grape varieties (e.g., Chardonnay, Traminette, and Vidal Blanc). However, large populations of beetles can also cause considerable leaf injury to lesser preferred varieties such as Concord (Figure 5).

Figure 4: Traminette leaves skeletonized by Japanese Beetle. Photo by: Andy Muza

Figure 5: Concord leaf with feeding injury from Japanese Beetle. Photo by: Andy Muza

Management

A presentation "Managing Japanese beetles in vineyards" (Rufuus Isaacs) (<http://www.isaacslab.ent.msu.edu/Images/talks/Isaacs%20Viticulture%202010%20JB%20for%20web.pdf>) provides information on various management strategies, scouting, leaf area loss tolerance and insecticide option information. Definitely worth checking out.

Biological and cultural management tactics are also discussed in various fact sheets about Japanese beetle but currently the most practical management in commercial vineyards requires insecticide application(s) aimed at adult beetles.

There are numerous insecticides registered on grapes for management of Japanese beetle in Pennsylvania, which can be found in the [2016 New York and Pennsylvania Pest Management Guidelines for Grapes](#). For a concise explanation of insecticide options refer to "Managing Japanese beetles in fruit crops" (R. Isaacs and J. Wise) (http://msue.anr.msu.edu/news/managing_japanese_beetles_in_fruit_crops).

Research has shown that grapevines can tolerate a fair amount of leaf area loss without detrimental effects. However, no economic threshold level has been established for leaf injury on grapes caused by Japanese beetle. Therefore, growers have to rely on their judgement and experience to determine leaf injury levels they can tolerate.

Before deciding if an insecticide application is needed in any of your vineyard blocks consider:

- Japanese beetle population levels,
- varietal susceptibility,
- age of vineyard (i.e., young or mature),
- canopy size, and
- crop load.

Heavy infestations in vineyards may require more than 1 insecticide application so frequent and thorough scouting of vineyards is necessary throughout the season. Many wine varieties, young vineyard blocks and vines in grow tubes are especially vulnerable to serious leaf loss by Japanese beetle feeding so consistent monitoring is critical.

Resources:

Japanese beetle, *Popillia japonica* Newman on Grape. (D.G. Pfeiffer and P.B. Schultz) <http://www.virginiafruit.ento.vt.edu/JBGrape.html>

Managing Japanese beetles in fruit crops (R. Isaacs and J. Wise) http://msue.anr.msu.edu/news/managing_japanese_beetles_in_fruit_crops

Managing Japanese beetles in vineyards (R. Isaacs) <http://www.isaacslab.ent.msu.edu/Images/talks/Isaacs%20Viticulture%202010%20JB%20for%20web.pdf>

Managing the Japanese Beetle: A Homeowner's Handbook https://www.aphis.usda.gov/publications/plant_health/2015/japanese-beetle-handbook.pdf

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5 responses to “Japanese Beetle: A Common Pest in the Vineyard”

Mark Chevalier says : July 11, 2016 at 3:20 am
Good article. The photos are helpful. Keep this kind of timely information coming.

REPLY

psuenology says : July 12, 2016 at 10:25 pm
Thanks for the support and continuing to follow us, Mark.

REPLY

Philip Jensen says : July 11, 2016 at 1:38 pm
I have observed that Japanese beetles do not like to fly at night and will stay "roosted" on the plants they are on making them easy targets for spray. Try to spray during the day, and they fly off, greatly reducing the effectiveness of contact type insecticides

REPLY

psuenology says : July 12, 2016 at 10:25 pm
Thank you for this information, Philip!

REPLY

Dean Scott says : July 12, 2016 at 2:52 am
Has anyone tried Milky Spore to eliminate Japanese Beetle larva in soil?

REPLY

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